FUTURE INTERNS / CYBER SECURITY

PHISHING EMAIL DETECTION & AWARENESS SYSTEM

A practical email threat analysis and user awareness report

PROJECT 02 • CYBER SECURITY INTERNSHIP

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PROJECT REPOSITORY
FUTURE_CS_02

TRAINING PROJECT • ALL EMAIL SAMPLES ARE CLEARLY IDENTIFIED AS SIMULATED

FUTURE INTERNS / CYBER SECURITY PROJECT 02

EXECUTIVE SUMMARY

Phishing Email Detection & Awareness System

PROJECT PURPOSE
Assess email content for common phishing indicators and translate the findings into practical guidance that helps users avoid deceptive requests.

01 PROJECT OVERVIEW

Phishing is a form of social engineering in which deceptive messages impersonate trusted people, organisations or services. The recipient is persuaded to disclose information, make a payment, visit a fraudulent website or take another unsafe action.

02 PROJECT OBJECTIVES

- Identify sender impersonation, look-alike domains, deceptive links and pressure tactics.
- Classify five training examples as Safe-style control, Suspicious or Phishing, based on the evidence available.
- Explain common phishing techniques in clear language.
- Describe potential risks and recommend safe user actions.
- Create a documented awareness report suitable for training and review.

03 METHOD AT A GLANCE

INPUT	REVIEW	OUTPUT
Five original simulated email examples	Sender, subject, wording, request, urgency and visible indicators	Classification, rationale, limitation and recommended action

04 EXPECTED OUTCOME

A structured report with five case studies, a comparative risk assessment, user-awareness guidance and references. The work is intended to be documented in the FUTURE_CS_02 repository.

Integrity note: all five email examples are fictional training scenarios. No live email, URL, attachment, sender infrastructure or malware analysis was performed.

FOUNDATIONS

INTRODUCTION TO PHISHING

Understanding the threat, its purpose and the human factors it exploits.

WHAT IS PHISHING?

Phishing is asocial-engineering technique in which an attacker impersonates a trusted person, organisation or service to influence a recipient into taking an unsafe action. This may involve disclosing credentials, sharing financial information, making a payment, opening an unexpected attachment or visiting a deceptive website.

CORE IDEA
Phishing targets both technology and human decision-making. A convincing message tries to make an unsafe action feel routine, urgent or trustworthy.

WHY EMAIL IS AN EFFECTIVE CHANNEL

Emailsupportseverydayactivitiessuchasinvoices,account notifications, delivery updates and password alerts. Attackers imitate these familiar workflows. A message can be grammatically correct and visually polished while still being deceptive.

A TYPICAL PHISHING SEQUENCE

STAGE	WHAT HAPPENS
1. Impersonation	The sender claims to represent a trusted organisation, colleague or service.
2. Persuasion	The message introduces urgency, fear, curiosity, a payment issue or an attractive offer.
3. Requested action	The recipient is encouraged to click, reply, pay, download or disclose information.
4. Potential impact	Credentials, money, personal data or account access may be exposed.

KEY PRINCIPLE

Assess therequest,not just the appearance of the message. Verify unexpected instructions through a separate, trusted channel before acting. [1–3]

THREAT LANDSCAPE

COMMON TYPES OF PHISHING

Different delivery methods share the same core tactic: impersonation and manipulation.

TYPE	DESCRIPTION
Email phishing	A broad message that imitates a familiar service or organisation.
Spear phishing	A targeted message tailored to a particular person or role.
Business Email Compromise (BEC)	A fraud attempt that impersonates an executive, employee or supplier to influence payments or sensitive business actions.
Whaling	A targeted attempt aimed at senior leaders or other high-authority individuals.
Smishing	A phishing attempt delivered through SMS or text messaging.
Vishing	A phishing attempt delivered through voice calls or voice-based communication.
Clone phishing	A copied message in which a link or attachment is replaced with a deceptive one.

WHY THE DISTINCTION MATTERS

Thechannelandtarget maydiffer,butthesamereviewhabits remain useful: confirm the sender independently, question unexpected requests, avoid opening unexpected links or attachments, and report suspicious messages through an approved process.

BUSINESS CONTEXT
BEC can involve realistic requests about invoices, supplier bank details, payroll or urgent transfers. Significant financial requests should be verified through a separate, established channel. [4]

ANALYST CHECKLIST

PHISHING INDICATORS

Indicatorsarecluestoinvestigate.Nosingleclueprovesthatamessage is malicious.

INDICATOR	WHAT TO LOOK FOR
Sender mismatch	The display name and actual email address do not match, or the address uses an unexpected domain.
Look-alike domain	A domain resembles a real organisation but includes added words, substituted characters or an unfamiliar ending.
Unexpected link	Visible link text and its destination may not match, or the destination may be unrelated to the claimed sender.
Urgency or threat	The message pressures the recipient to act immediately or threatens account closure, penalties or loss.
Sensitive-data request	The email asks for passwords, one-time codes, PINs, payment details or identity information.
Unexpected attachment	The message asks the recipient to open an invoice, document or archive that was not expected.
Unusual payment instruction	A supplier or colleague requests changed bank details, gift cards, a wire transfer or an unusual payment route.
Context mismatch	The recipient has no known relationship with the sender, transaction, delivery or service mentioned.

HOW TO INTERPRET THE CLUES

Genericgreetingsand spellingerrorscan bewarning signs, but their absence does not make a message safe. An unfamiliar sender is not automatically a phisher. Evaluate multiple indicators together and seek independent verification. [1–3]

ANALYST REMINDER
A message can contain a suspicious request even when it has no spelling mistakes, and a legitimate message can occasionally look unusual. Context and verification matter.

PROJECT METHOD

SCOPE & METHODOLOGY

A transparent, limited-scope review of five simulated email examples.

PROJECT SCOPE

Thisreportdemonstratesarepeatable first-line review of email content. It focuses on visible sender details, subject lines, message wording, requested actions, urgency and any stated link or attachment indicators.

DATASET DESCRIPTION

Thedataset consistsof five original,fictional training examples written for this project. They are not intercepted emails, are not claimed to come from a public dataset and do not represent actual victims or organisations. Reserved example domains are used; no live malicious links or attachments are included.

REVIEW PROCEDURE

STEP	ACTION
1. Record	Capture the sample ID, sender string, subject and message text.
2. Inspect	Identify the requested action, urgency, sensitive-data requests, and link or attachment references.
3. Compare	Check whether the sender and request fit the expected context; note mismatches and impersonation clues.
4. Classify	Assign a category and document the supporting reasoning and confidence limits.
5. Recommend	State the safest next step and identify what further evidence would be needed.

LIMITATIONS & ETHICAL HANDLING

Nooriginal emailfiles, fullheaders,authenticationresults, DNS records, URL destinations or attachments were supplied. SPF, DKIM, DMARC, sender ownership, URL reputation and malware status were therefore not tested. The classifications are educational judgments based on simulated content, not forensic findings. Real investigations should follow organisational policy and use approved tools in a controlled environment.

EMAIL CASE STUDY / E01

ROUTINE ACCOUNT NOTIFICATION

SIMULATED TRAINING SAMPLE • CONTENT CREATED FOR THIS PROJECT

CLASSIFICATION: SAFE-STYLE CONTROL

FROM	notifications@service.example
TO	user@example.com (fictional)
SUBJECT	Your monthly account statement is available
DATE	Fictional training message

MESSAGE CONTENT

Dear Customer,

Your monthly account statement is now available for review. You can access your statement by opening the official service application or by visiting the website address you normally use.

For your security, never share your password or verification codes with anyone.

Thank you,
Customer Support Team

OBSERVABLE INDICATORS

FIELD	OBSERVATION
Sender	Reserved .example domain; illustrative, not a real sender.
Request	No password, payment or personal-information request.
Link / attachment	No embedded link or attachment is included.
Pressure	No deadline or threat is used.

ANALYST ASSESSMENT

Thesimulatedcontentcontainsnoobvious phishing request or pressure tactic. It is included as a benign-style comparison sample.

RECOMMENDED USER ACTION

Treat this only as a safe-style control for the exercise. In real use, verify the sender and message context; visible text alone cannot authenticate an email.

EVIDENCE LIMITATION

No real message headers or sender-authentication evidence exist. This example is not a verified legitimate email.

EMAIL CASE STUDY / E02

UNEXPECTED INVOICE NOTIFICATION

SIMULATED TRAINING SAMPLE • CONTENT CREATED FOR THIS PROJECT

CLASSIFICATION: SUSPICIOUS

FROM	billing@unknown-vendor.example
TO	user@example.com (fictional)
SUBJECT	Invoice pending — payment confirmation required
DATE	Fictional training message

MESSAGE CONTENT

Dear Customer,

We have generated a new invoice associated with your account. Please review the attached invoice and confirm the payment status.

If you do not recognise this transaction, contact the billing department through an independently verified channel.

Regards,
Billing Department

OBSERVABLE INDICATORS

FIELD	OBSERVATION
Sender	Unknown vendor-style address; authenticity is not established.
Request	Asks the recipient to review an unexpected invoice and confirm payment status.
Attachment	An attachment is mentioned but none is supplied in this simulation.
Context	No existing vendor relationship is established in the example.

ANALYST ASSESSMENT

Anunexpected invoiceand unfamiliarsender justify caution. The message does not provide enough evidence to conclude that it is definitely phishing.

RECOMMENDED USER ACTION

Do not open an unexpected attachment. Check purchase records and contact the vendor using independently sourced contact details.

EVIDENCE LIMITATION

No attachment, headers or vendor records were provided; sender authenticity and file risk remain untested.

BANK ACCOUNT VERIFICATION REQUEST

SIMULATED TRAINING SAMPLE • CONTENT CREATED FOR THIS PROJECT

CLASSIFICATION: PHISHING

FROM	security-alert@bank-verification.example
TO	user@example.com (fictional)
SUBJECT	Immediate action required: verify your bank account
DATE	Fictional training message

MESSAGE CONTENT

Dear Customer,

We detected an issue with your account information. To prevent temporary restrictions, reply with your account details and one-time verification code.

Please complete this request immediately.

Security Team

OBSERVABLE INDICATORS

FIELD	OBSERVATION
Sender	Fictional, unverified bank-style domain.
Sensitive data	Requests account details and a one-time verification code.
Urgency	Uses immediate-action and account-restriction pressure.
Channel	Asks the recipient to disclose sensitive information by reply.

ANALYST ASSESSMENT

Thedirectrequest foraccountdetailsanda one-time code, combined with urgency and an unverified sender, is a strong phishing pattern.

RECOMMENDED USER ACTION

Do not reply or disclose codes. Contact the bank through its official app, card number or independently typed website; report the message.

EVIDENCE LIMITATION

Fictional training example. No real bank, sender infrastructure or email headers were examined.

EMAIL CASE STUDY / E04

PARCEL DELIVERY PAYMENT REQUEST

SIMULATED TRAINING SAMPLE • CONTENT CREATED FOR THIS PROJECT

CLASSIFICATION: PHISHING

FROM	delivery-update@parcel-notice.example
TO	user@example.com (fictional)
SUBJECT	Delivery on hold — small fee required
DATE	Fictional training message

MESSAGE CONTENT

Hello,

Your parcel could not be delivered because a handling fee is pending. Use the payment option in this message to release the shipment today.

If payment is not completed, the delivery may be returned.

Delivery Support

OBSERVABLE INDICATORS

FIELD	OBSERVATION
Sender	Unverified parcel-notice domain.
Payment	Requests an unexpected fee to release a parcel.
Urgency	Creates pressure through a same-day deadline and threat of return.
Link	Refers to a payment option but contains no live URL.

ANALYST ASSESSMENT

Anunexpected delivery feepairedwithurgency and an unverified sender is consistent with common parcel-delivery phishing lures.

RECOMMENDED USER ACTION

Do not use the message's payment route. Check tracking directly through the courier's official app or website using a tracking number from the original purchase.

EVIDENCE LIMITATION

No real tracking number, payment page, headers or courier account were supplied or tested.

EMAIL CASE STUDY / E05

URGENT PASSWORD RESET NOTIFICATION

SIMULATED TRAINING SAMPLE • CONTENT CREATED FOR THIS PROJECT

CLASSIFICATION: PHISHING

FROM	account-security@password-help.example
TO	user@example.com (fictional)
SUBJECT	Your password expires today — reset immediately
DATE	Fictional training message

MESSAGE CONTENT

Dear User,

Your account password is scheduled to expire today. To avoid losing access, confirm your current password and verification code through the password reset page referenced in this notice.

Failure to complete the process may result in account suspension.

Account Security

OBSERVABLE INDICATORS

FIELD	OBSERVATION
Sender	Unverified account-security style domain.
Credentials	Requests the current password and a verification code.
Urgency	Threatens loss of access and demands immediate action.
Link	A reset page is referenced but no destination is supplied for testing.

ANALYST ASSESSMENT

Arequest toprovideanexistingpasswordand verification code through an unsolicited message is a strong phishing indicator.

RECOMMENDED USER ACTION

Do not enter credentials through the message. Open the service using a known bookmark or official app, check account security notifications, and report the email.

EVIDENCE LIMITATION

No live URL, service identity, email headers or authentication records were available for technical verification.

COMPARATIVE RESULTS

CLASSIFICATION & RISK ASSESSMENT

Summary of the five simulated examples and the reasoning behind each label.

ID	SCENARIO	LABEL	BASIS	POTENTIAL RISK
E01	Routine account notification	**Safe-style control***	No obvious unsafe request in simulated text.	Low observable concern
E02	Unexpected invoice	**Suspicious**	Unknown sender and unexpected invoice context; insufficient proof.	Needs verification
E03	Bank verification request	**Phishing**	Requests account details and one-time code under urgency.	High potential impact
E04	Parcel delivery fee	**Phishing**	Unexpected payment request and deadline pressure.	Medium–high potential impact
E05	Urgent password reset	**Phishing**	Requests current password and verification code.	High potential impact

HOW TO READ THE RESULTS

Theriskdescriptions refertopotential impactifarecipient followed the request. They are not measured probabilities and are not results from a live security tool. A real assessment would also consider message authenticity, recipient context, destination reputation, attachment behaviour and organisational controls.

CLASSIFICATION DEFINITIONS

- **Safe-stylecontrol:**a benign-lookingcomparisonmessage. This label does not certify sender authenticity.
- **Suspicious:** warning signs are present, but evidence is insufficient for a definitive phishing conclusion.
- **Phishing:** the simulated message contains strong deceptive indicators, such as credential or one-time-code harvesting.

IMPORTANT QUALIFICATION
E01 is not a verified safe email. All five samples are fictional. No real message headers, sender infrastructure, URLs or attachments were tested.

USER SECURITY

PREVENTION & AWARENESS GUIDELINES

BEFORE ACTING ON AN EMAIL

- Pause: do notle turgency,fear oran attractiveoffer rush your decision.
- Check the sender: inspect the full email address, not only the display name.
- Verify independently: use a known phone number, official app or bookmark—not contact details supplied in the unexpected message.
- Protect credentials: never disclose passwords, PINs or one-time verification codes in response to an unsolicited email.
- Review links carefully: do not open unexpected links; follow approved organisational procedures if a check is required.
- Treat attachments cautiously: confirm that a file is expected and comes from a trusted source before opening it.
- Verify payment changes: confirm supplier bank-detail changes or urgent transfers through a separate, established channel.
- Report suspicious messages: use the provider's phishing-report function or the organisation's security team.

IF A MESSAGE MAY BE PHISHING

Donotreply,pay,discloseinformation,oropenitslinksor attachments. Preserve the message if the organisation's incident process requires it, report it through the approved channel, and follow the security team's instructions. If credentials may have been disclosed, change them through the official service and contact the provider or IT team promptly.

ACCOUNT PROTECTION HABITS

PRACTICE	WHYIT HELPS
Unique passwords	Reduces the risk that one exposed password unlocks several accounts.
Multi-factor authentication	Adds another verification step; phishing-resistant methods are preferable where available.
Software updates	Helps address known security weaknesses in devices and applications.

Guidance informed by public recommendations from the FTC, FBI, CISA and Google. [1–5]

CLOSING ASSESSMENT

RECOMMENDATIONS & CONCLUSION

RECOMMENDATIONS FOR INDIVIDUALS

- Useapause-and-verifyhabitwhenever amessage asksformoney,credentials or urgent action.
- Accessservices through a trusted app, bookmark or known address rather than an unexpected email link.
- Reportsuspected phishing using the provider's or organisation's reporting process.
- If an account may be compromised, contact the service provider and follow its official recovery process.

RECOMMENDATIONS FOR ORGANISATIONS

- Provide short,recurringawarenesstraining using realistic but controlledexamples.
- Publish a simple reporting route and make it easy for staff to flag suspicious messages.
- Require out-of-band verification for payment changes, bank-detail updates and unusual sensitive requests.
- Deploy and monitor SPF, DKIM and DMARC alongside filtering and account protections.
- Maintain an incident-response process for suspected credential disclosure, malware delivery or payment fraud.

CON CLUSION

Thisproject demonstrates a structured method for reviewing email content, identifying common phishing indicators and communicating risk in plain language. The five simulated examples show how sender details, context, requested actions and urgency can influence classification. They also demonstrate why a Suspicious category is useful when evidence is incomplete.

KEY TAKEAWAY
Verify unexpected requests independently. Protect passwords and verification codes. Email appearance alone cannot establish authenticity; stronger conclusions require original headers and supporting technical evidence.

PROJECT LIMITATION & INTEGRITY STATEMENT

All five email examples are fictional training scenarios created for this report. No real users, organisations, malicious links or attachments were involved. No live email-header, domain, URL or malware analysis was performed. Findings are educational and must not be represented as results of a live investigation.

SOURCE MATERIAL

REFERENCES

Publicguidanceusedtoinformtheconcepts and user-protection recommendations in this report.

[1] Federal Trade Commission (FTC)
How To Recognize and Avoid Phishing Scams.

https://consumer.ftc.gov/articles/how-recognize-avoid-phishing-scams

[2] Google Account Help
Avoid & report phishing emails.

https://support.google.com/mail/answer/8253

[3] Federal Bureau of Investigation (FBI)
Spoofing and Phishing.

https://www.fbi.gov/how-we-can-help-you/common-frauds-and-scams/spoofing-and-phishing

[4] Federal Bureau of Investigation (FBI)
Business Email Compromise.

https://www.fbi.gov/how-we-can-help-you/common-frauds-and-scams/business-email-compromise

[5] Cybersecurity and Infrastructure Security Agency (CISA)
Phishing: Simple Tips (security awareness guidance).

https://www.cisa.gov/sites/default/files/publications/Phishing%20General%20Security%20Postcard_6.24.2021_508cV2.pdf

[6] Future Interns
CyberS ecurityInternship Guidelines and Task Descriptions, supplied project brief.

User-provided project brief; requirements referenced for project framing.

REFERENCE USE & SCOPE

The sources above support general definitions, common tactics and user-protection advice. The email samples and classifications are original simulated material prepared for this project; they are not quoted from these sources. The Future Interns brief is the source for the internship task framing. Reference access date: 02 November 2026.

END OF REPORT
FUTURE_CS_02 • PHISHING EMAIL DETECTION & AWARENESS SYSTEM